1789. Primary Department for Each Employee

PLATFORM Platform: LeetCode
DIFFICULTY EASY
DATE Solved: June 22, 2026

Input

- 1 table — Employee with columns employee_id, department_id, primary_flag
- primary_flag = 'Y' → primary department (only set when employee has multiple departments)
- primary_flag = 'N' → either not primary OR employee has only 1 department

Output

- Each employee with their primary department

Key Insights

- 2 separate cases exist → need 2 queries combined with UNION
- Single department employees have primary_flag = 'N' — can't filter by flag alone
- UNION removes duplicates automatically

Observations

- Employee with 1 department → primary_flag is always 'N'
- Employee with multiple departments → exactly one has primary_flag = 'Y'
- Both queries must return the same columns for UNION to work

Approach

- Case 1 → Multiple department employees → WHERE primary_flag = 'Y'
- Case 2 → Single department employees → GROUP BY + HAVING COUNT = 1
- Combine → UNION

Algorithm

Query 1: WHERE primary_flag = 'Y'
→ gives multi-department employees' primary dept

UNION

Query 2: GROUP BY employee_id
HAVING COUNT(department_id) = 1
→ gives single-department employees

Final Result → all employees covered

Section

	Complexity
TC	O(N) — scan table twice for 2 queries
SC	O(N) — storing results of both queries

Points to Remember

- UNION → combines results, removes duplicates
- Both queries need same number/type of columns
- When problem has 2 different logical cases → think UNION
- Don't filter by flag alone — single-department employees won't have 'Y'

Pattern

UNION Pattern for Multiple Cases

Whenever you see:

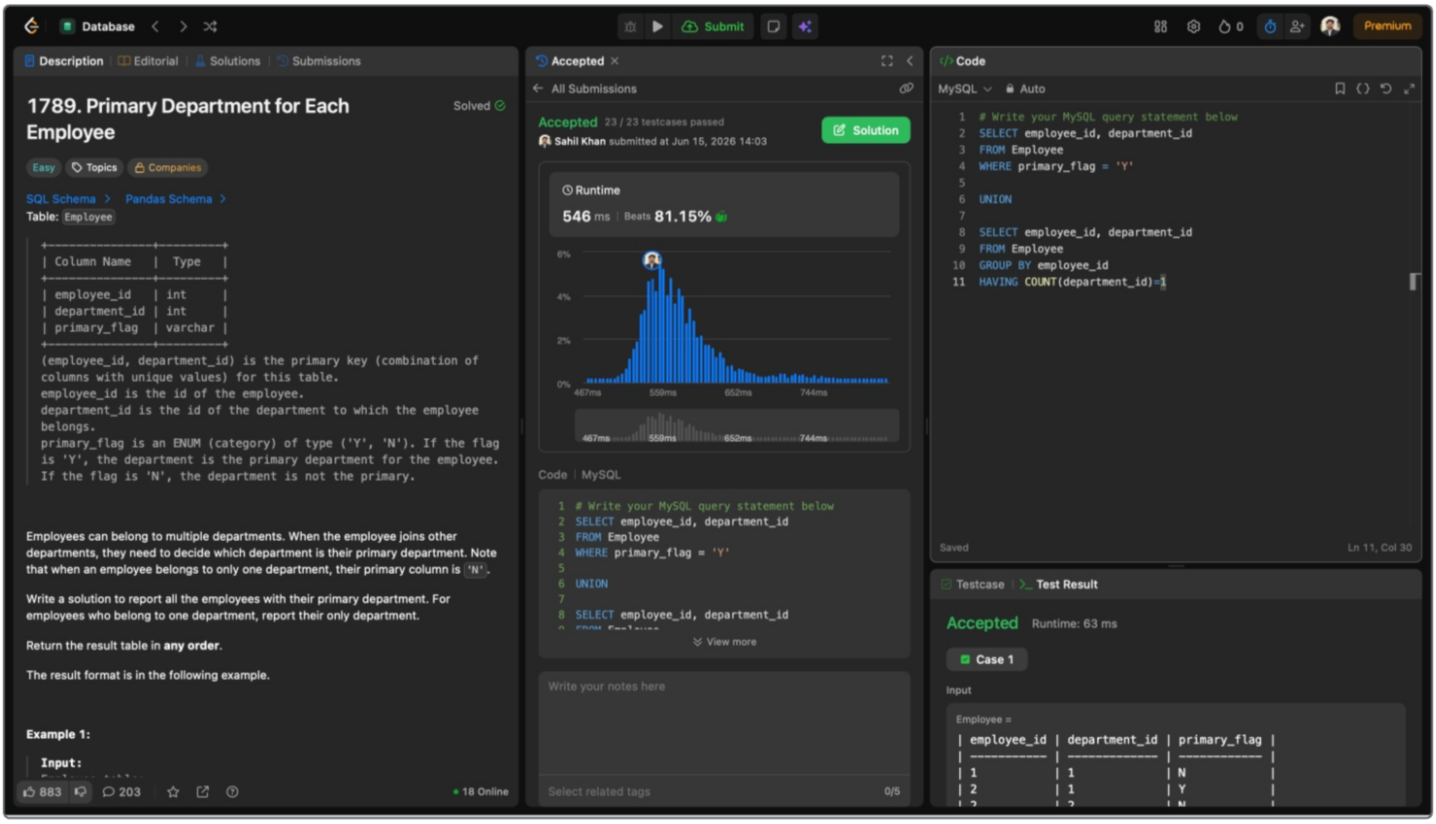
- "Two different conditions to pick rows"
- "Combine results from different scenarios"

Think →

Query for Case 1

UNION

Query for Case 2


Database
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1789. Primary Department for Each Employee
Solved
Easy
Topics
Companies
SQL Schema
Pandas Schema
Table: Employee
+---------------+---------+
| Column Name | Type |
+---------------+---------+
employee_id	int
department_id	int
primary_flag	varchar
+---------------+---------+
(employee_id, department_id) is the primary key (combination of columns with unique values) for this table.
employee_id is the id of the employee.
department_id is the id of the department to which the employee belongs.
primary_flag is an ENUM (category) of type ('Y', 'N'). If the flag is 'Y', the department is the primary department for the employee. If the flag is 'N', the department is not the primary.
Employees can belong to multiple departments. When the employee joins other departments, they need to decide which department is their primary department. Note that when an employee belongs to only one department, their primary column is 'N'.
Write a solution to report all the employees with their primary department. For employees who belong to one department, report their only department.
Return the result table in any order.
The result format is in the following example.
Example 1:
Input:
883
203
18 Online
Accepted
All Submissions
Accepted 23 / 23 testcases passed
Sahil Khan submitted at Jun 15, 2026 14:03
Solution
Runtime
546 ms | Beats 81.15%
Code | MySQL
View more
Write your notes here
Select related tags 0/5
Code
MySQL Auto
Write your MySQL query statement below
SELECT employee_id, department_id
FROM Employee
WHERE primary_flag = 'Y'
UNION
SELECT employee_id, department_id
FROM Employee
GROUP BY employee_id
HAVING COUNT(department_id)=1
Saved
Ln 11, Col 30
Testcase | Test Result
Accepted Runtime: 63 ms
Case 1
Input
Employee =
employee_id	department_id	primary_flag
1	1	N
2	1	Y